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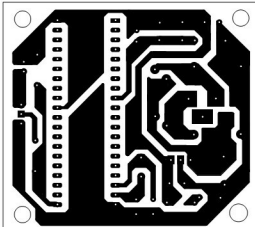


Fig. 4: Actual-size, single-side PCB for the microcontroller-based ring tone generator

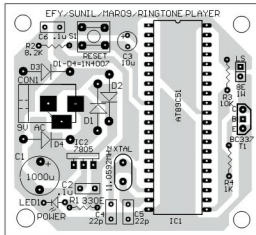


Fig. 5: Component layout for the PCB

(P2.0 = 1) and 'low' (P2.0 = 0), i.e., the bit remains 'high' for a certain period of time and 'low' for the same period of time (refer Fig. 2).

Time period of the note (Ts) = 1/frequency of note

Therefore,

Duration of 'high' bit (time for which P2.0 is '1') = $T_s/2 = t_{s1}$ (half-time period)

Duration of 'low' bit (time for which P2.0 is '0') = $T_s/2 = t_{s1}$ (half-time period)

So you need to use two timers for duration D and half-time period 'ts.' Use one timer to generate 1's and 0's at a time interval of 'ts' and the other timer to interrupt the PWM signal after time interval D.

We have used two timers for time-

period calculation as follows:

1. Timer 0 for calculating the duration (D)

2. Timer 1 for calculating the half time period (ts)

The CPU takes certain number of clock cycles to execute an instruction. The simplest instruction takes a single byte of code and executes in one machine cycle. The standard 8051 machine cycle is equal to twelve oscillator cycles. We have used a 11.0592MHz crystal.

So time period = $1/(11.0592 \times 10^6) = 0.0904 \mu s$

Time period of a machine cycle = $0.0905 \times 12 = 1.085 \mu s$

Timer 0. Timer 0 is a 16-bit timer that is used for duration 'D'. It is loaded with value DC00H. Therefore the number of machine cycles taken by the timer before it is reset = $FFFF - DC00 + 1 = 2400H = 9216$ in decimal.

Therefore time taken 't' = $9216 \times 1.085 \mu s = 0.001$ second

So after timer 0 is set, it will take 0.001 second to reset.

For 1-second duration, the timer needs to be set $1/0.001 = 100$ times.

For duration 'D', timer 0 needs to be set Dx100 times.

Timer 1. The half time period is 'ts.'

Therefore the number of machine cycles needed = $t_{s1}/1.085 \mu s$

If the value of the timer is 'x', then $FFFF - x + 1 = t_{s1}/(1.085 \times 10^{-6})$

From this equation, the value of 'x' can be calculated.

1's and 0's are continuously generated from port pin P2.0 at an interval of 'ts' seconds alternatively until duration 'D' (in seconds) completes (refer Table II).

Circuit description

Fig. 3 shows the circuit of the microcontroller-based ring tone generator.